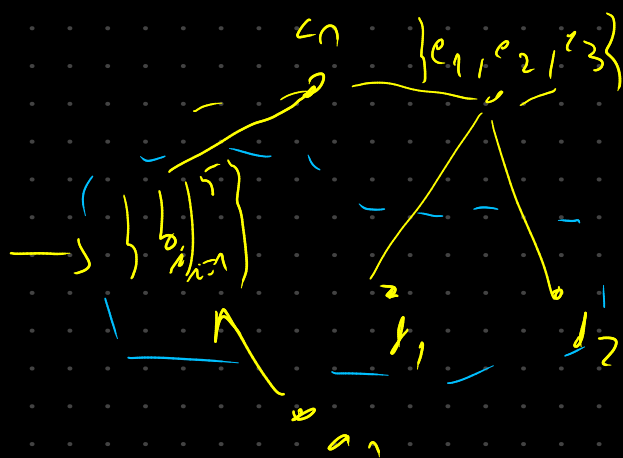
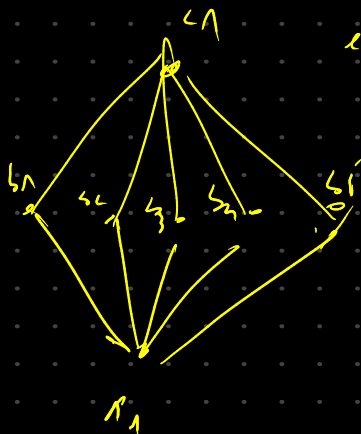
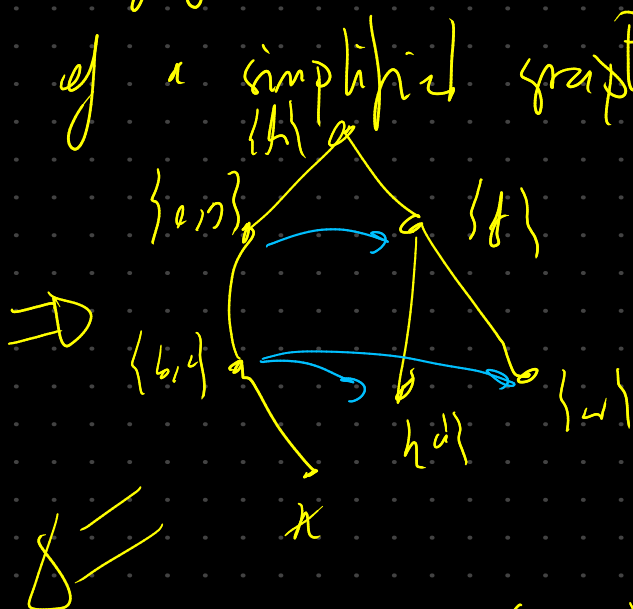
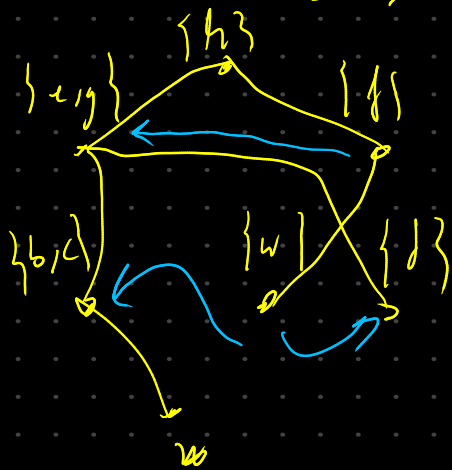


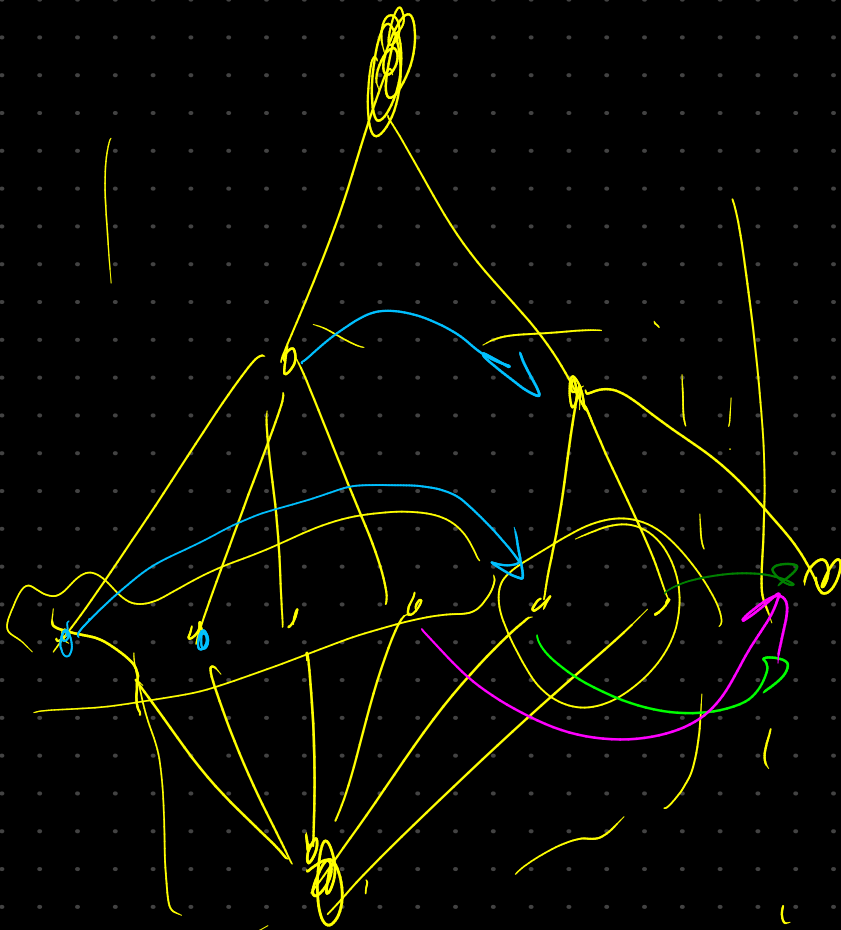
objectif : - greedy strategy

- use of a simplified graph



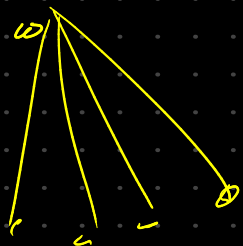
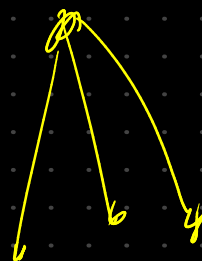
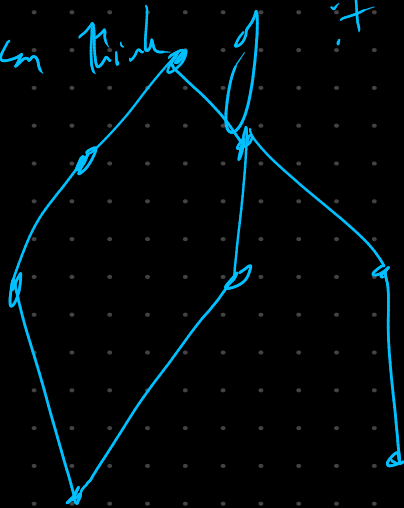
order of the
connected comp.
during construction

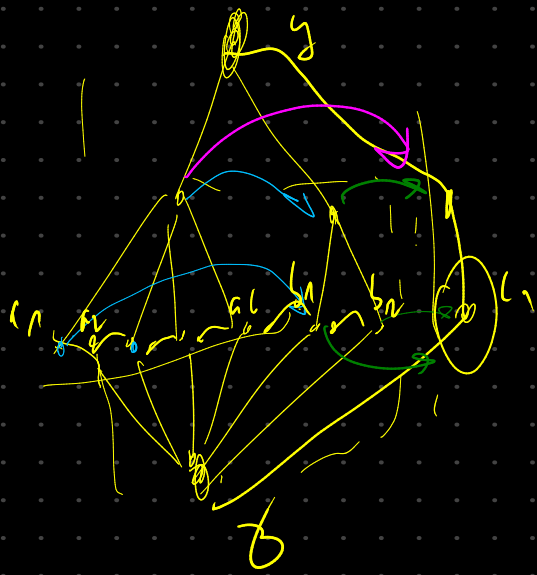
no cycles
arbitrary non cyclic order!



green
map
she is in $\mathcal{N} \cup \mathcal{E}$

vertices rel. node that are not rep. by 2-sets
are also the vertex that have the same Neighbors.
(we can think of it as a single edge)





$$\begin{aligned} \varphi(a_1 b_1) &= \varphi(a_1 b_1) \cdot 1 \\ &= \varphi(a_2) \cdot \dots \\ &= \varphi(a_3) \cdot \dots \\ &= \varphi(a_6) \cdot \dots \end{aligned}$$

$$\begin{pmatrix} a_1 & a_2 & a_3 & a_4 & a_5 & a_6 & c_1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ b_1 & b_2 & b_3 & b_4 & b_5 & b_6 & 0 \end{pmatrix}$$

$$\begin{pmatrix} x_1 & x_2 & x_3 & x_4 & x_5 & x_6 \\ x_1 & x_2 & x_3 & x_4 & x_5 & x_6 \end{pmatrix}$$

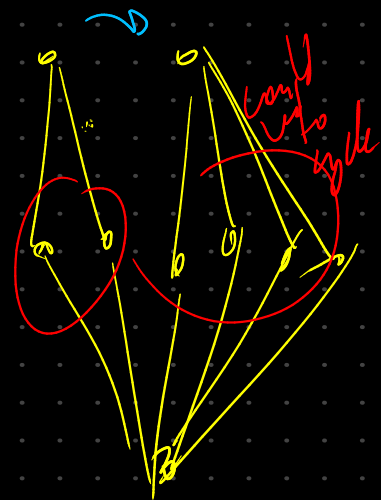
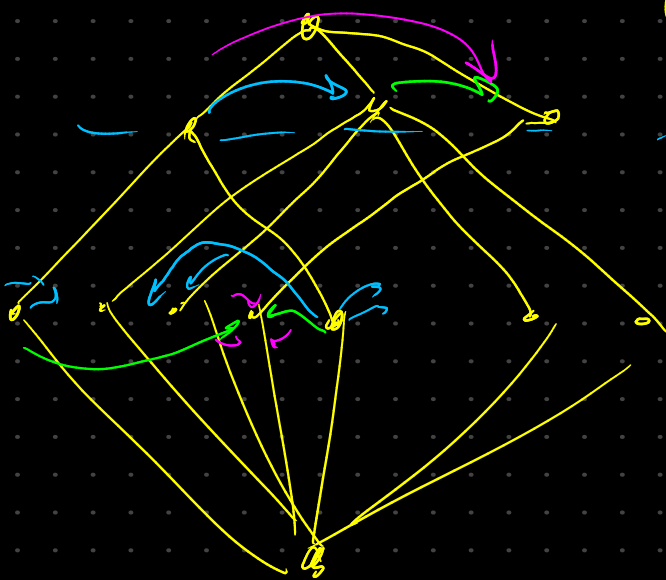
$$\begin{pmatrix} x_2 & x_3 & x_4 & x_5 & x_6 & 0 \end{pmatrix}$$

$$\varphi(x_1) = 2$$

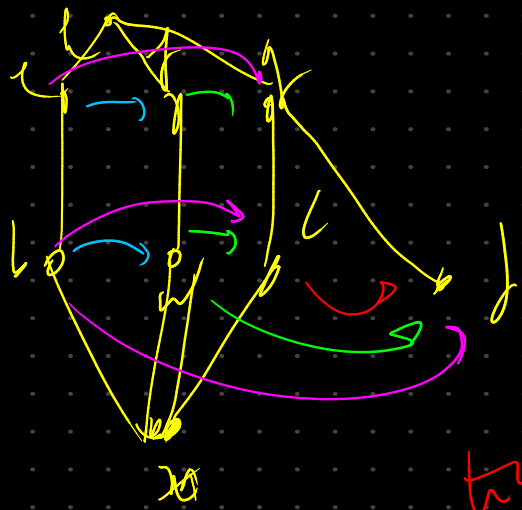
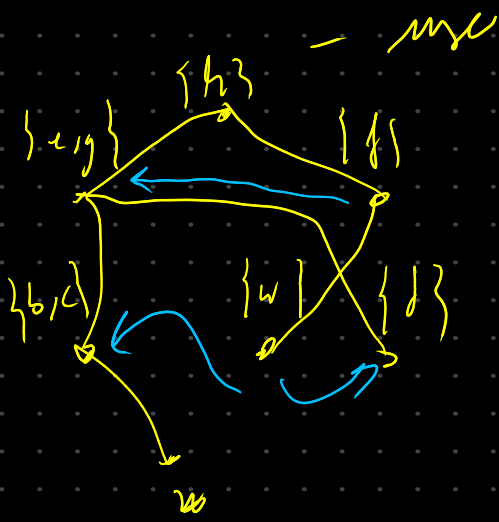
$$\varphi(a_1 b_1) = 1 \Rightarrow \varphi$$

no self crossing in
vertices that share
one vertex from level below
resolve drawing without cycles

cycles φ
line first

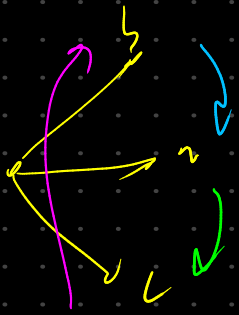


second
3



$$\varphi_n = 1 \quad \varphi_1 = 1 \quad \varphi_2 = 0 \Rightarrow \varphi_n = 0$$

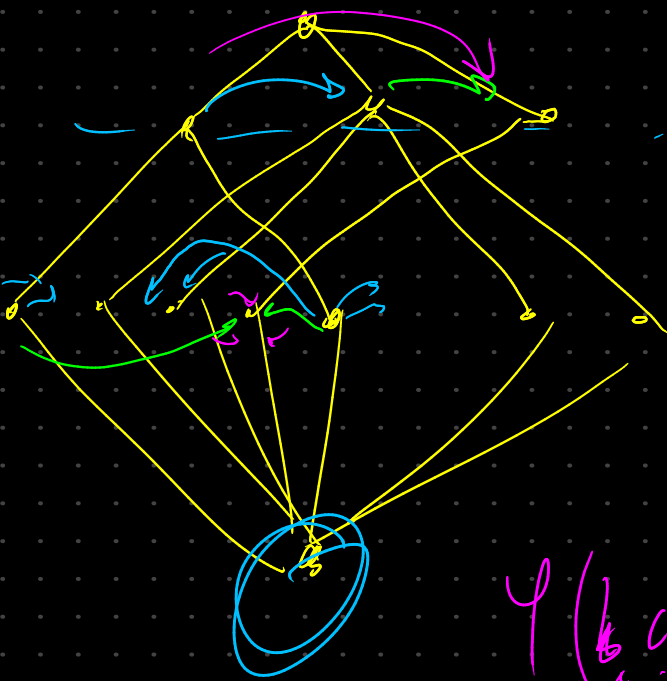
$$\varphi_n = 1 \quad \varphi_4 = 0$$



$$\begin{aligned} b_w &= 1 \\ w_c &= 1 \\ c_b &= 1 \\ b_w &= 1 \\ w_c &= 1 \\ c_b &= 1 \end{aligned}$$

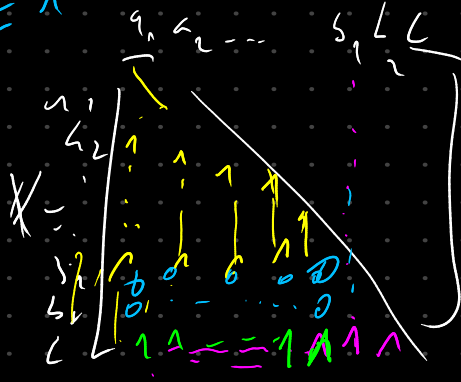


keep track of adjacent edges and their order

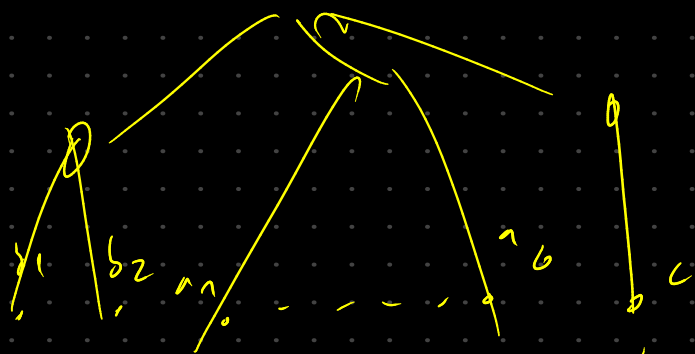


$$\varphi_n = 1$$

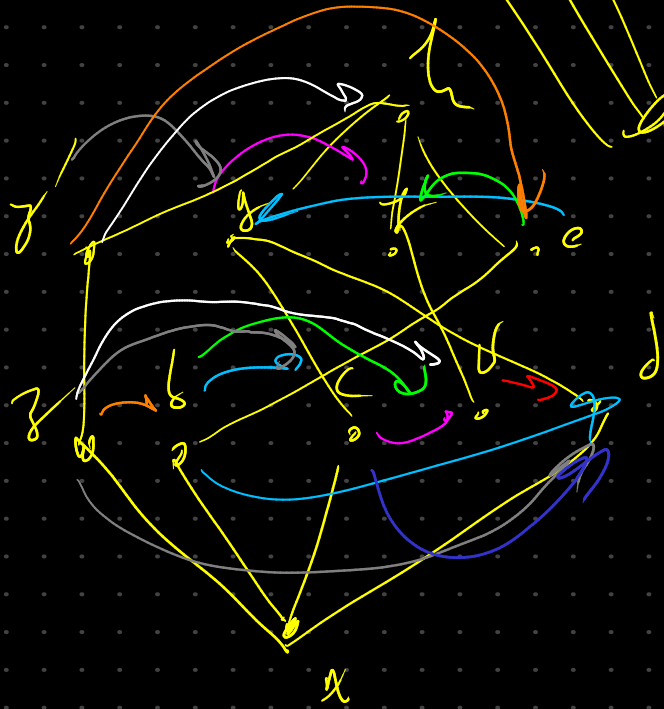
adjacency:



$$\varphi(b, c) \quad \varphi$$



' o c



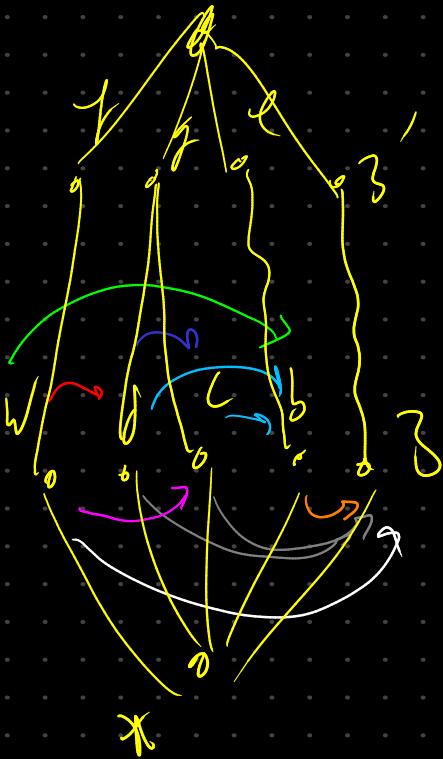
h l u s $\begin{matrix} p_1 & p_2 & p_3 & p_4 \\ \wedge & \wedge & \wedge & \wedge \\ o & o & o & o \end{matrix}$

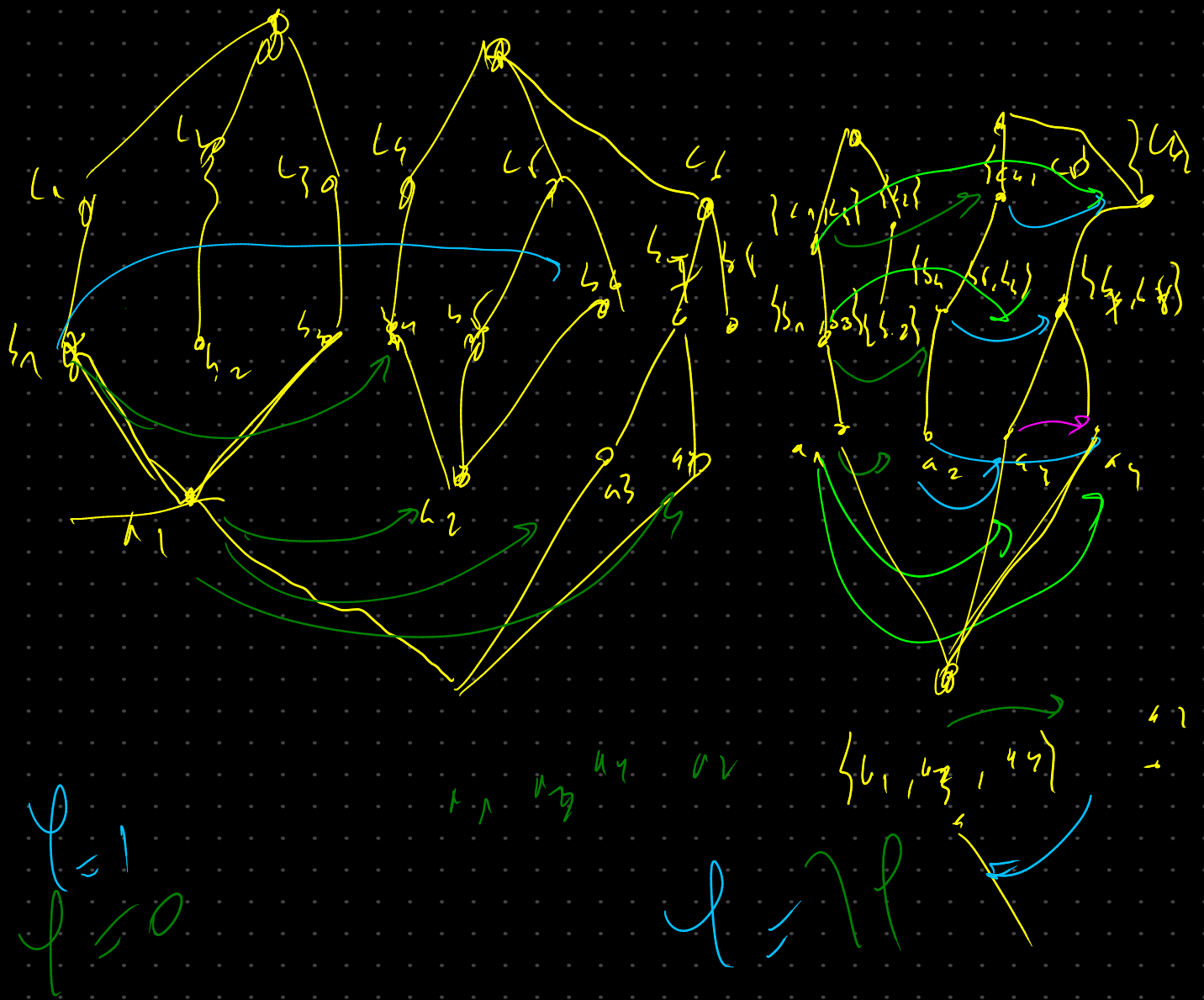
h

$\begin{matrix} p_1 & p_2 & p_3 & p_4 \\ \wedge & \wedge & \wedge & \wedge \\ o & o & o & o \end{matrix}$



$\begin{matrix} p_1 & p_2 & p_3 & p_4 \\ \wedge & \wedge & \wedge & \wedge \\ o & o & o & o \end{matrix}$





The idea for now is to go through level by level, assign ϕ 's existing in that level and then go to the next level, I still need to consider the conditions and write them in way it shows easily in the relative relation of 2-sat